

UNIVERSITÀ DEGLI STUDI DI TORINO

Corso di Laurea Magistrale in Matematica



UNIVERSITÀ
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Tesi di Laurea Magistrale

Teoria delle categorie

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Introduzione

Testo

Capitolo 1

Teoria

1.1 The pullback functor

In this section we will define the pullback functor and its left adjoint, the shrink functor.

Given a category $\mathcal{A}$ with pullbacks and a morphism $p : E \rightarrow A$ in $\mathcal{A}$, one can consider adjunction

$$(\mathcal{A} \downarrow B) \quad \begin{array}{c} \xrightarrow{p^*} \\ \perp \\ \xleftarrow{p!} \end{array} \quad (\mathcal{A} \downarrow E)$$

defined as follows:

- $(\mathcal{A} \downarrow B)$ is the category of objects over B in $\mathcal{A}$, we denote an object by $(A, \alpha : A \rightarrow B)$;
- $(\mathcal{A} \downarrow E)$ is the category of objects over E in $\mathcal{A}$, we denote an object by $(D, \delta : D \rightarrow E)$;
- $p^* : (\mathcal{A} \downarrow E) \rightarrow (\mathcal{A} \downarrow B)$ is the pullback functor

$$\begin{array}{ccc} \begin{array}{c} A \\ \downarrow \alpha \\ B \end{array} & \longmapsto & \begin{array}{ccccc} E \times_B A & \xrightarrow{\text{proj}_2} & A \\ \downarrow \text{proj}_1 & & \downarrow \alpha \\ E & \xrightarrow{p} & B \end{array} \end{array}$$

$$\begin{array}{ccc}
 \begin{array}{ccc} A & \xrightarrow{f} & A' \\ \downarrow \alpha & \nearrow \alpha' & \\ B & & \end{array} & \mapsto & \begin{array}{c} \begin{array}{ccccc} & & \exists! \tilde{f} & & \\ & \curvearrowright & \text{---} & \curvearrowleft & \\ E \times_B A & \rightarrow & A & \xrightarrow{f} & A' & \leftarrow & E \times_B A' \\ & \searrow \alpha & & \swarrow \alpha' & \\ & & B & & \\ \text{proj}_1 \swarrow & & \uparrow p & & \searrow \text{proj}'_1 \\ & & E & & \end{array} \end{array}
 \end{array}$$

- $p! : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)$ is the shrink functor

$$\begin{array}{ccc}
 \begin{array}{ccc} D & & \\ \downarrow \delta & & \\ E & & \end{array} & \mapsto & \begin{array}{ccc} D & & \\ \delta \downarrow & \searrow p\delta & \\ E & \xrightarrow{p} & B \end{array}
 \end{array}$$

$$\begin{array}{ccc}
 \begin{array}{ccc} D & \xrightarrow{d} & D' \\ \delta \downarrow & \nearrow \delta' & \\ E & & \end{array} & \mapsto & \begin{array}{ccc} D & \xrightarrow{d} & D' \\ \delta \downarrow & \searrow p\delta & \downarrow p\delta' \\ E & \xrightarrow{p} & B \end{array}
 \end{array}$$

- the unit η^p

$$\eta : \text{Id}_{(\mathcal{A} \downarrow E)} \rightarrow p^* p!$$

$$\eta_{(D, \delta)} = \langle \delta, 1_D \rangle : (D, \delta) \rightarrow (E \times_B D, \text{proj}_1)$$

$$\begin{array}{c}
 \begin{array}{ccccc}
 D & & & & \\
 \delta \searrow & & \xrightarrow{1_D} & & \\
 & & E \times_B D & \xrightarrow{\quad} & D \\
 & & \downarrow \text{proj}_1 & & \downarrow p\delta \\
 & & E & \xrightarrow{p} & B
 \end{array}
 \end{array}$$

- the counit ϵ^p

$$\epsilon : p_! p^* \rightarrow \text{Id}_{(A \downarrow B)}$$

$$\epsilon_{(A, \alpha)} = \text{proj}_2 : (E \times_B A, p(\text{proj}_1)) \rightarrow (A, \alpha)$$

$$\begin{array}{ccc} E \times_B A & \xrightarrow{\text{proj}_2} & A \\ \downarrow \text{proj}_1 & & \downarrow \alpha \\ E & \xrightarrow{p} & B. \end{array}$$

We can check that the unit and counit fulfil the triangle identities:

$$((p^* \epsilon) \circ (\eta p^*))_{(A, \alpha)} = (p^* \epsilon)_{(A, \alpha)} (\eta p^*)_{(A, \alpha)} = p^* (\text{proj}_2) (\eta_{(E \times_B A, \text{proj}_1)}) = \circledast$$

considering how p^* acts on the morphism $\text{proj}_{1, E \times_B A}$, we have

$$\begin{array}{ccccc} E \times_B A & \xrightarrow{\text{proj}_{1, E \times_B A}} & A & & \\ & \searrow \alpha(\text{proj}_{2, E \times_B A}) & \swarrow \alpha & & \\ & & B & & \end{array}$$

$$\begin{array}{ccccccc} & & \text{proj}_{2, E \times_B A} & & & & \\ & \text{proj}_{2, E \times_B A} & \xrightarrow{\quad} & \text{proj}_{2, E \times_B A} & \xrightarrow{\quad} & \text{proj}_{2, E \times_B A} & \\ E \times_B (E \times_B A) & \xrightarrow{\text{proj}_{2, E \times_B (E \times_B A)}} & E \times_B A & \xrightarrow{\text{proj}_{2, E \times_B A}} & A & \xleftarrow{\text{proj}_{2, E \times_B A}} & E \times_B A \\ & \searrow (\alpha)(\text{proj}_{2, E \times_B A}) & \searrow \alpha & & \swarrow \alpha & & \\ & & B & & & & \\ & \swarrow \text{proj}_{1, E \times_B (E \times_B A)} & \uparrow p & \swarrow \text{proj}_{1, E \times_B A} & & & \\ & & E & & & & \end{array}$$

$$\circledast = (\tilde{\text{proj}}_2) (\eta_{(E \times_B A, \text{proj}_1)})$$

then considering the definition of η

$$\begin{array}{ccccc}
 & & 1_{E \times_B A} & & \\
 & \nearrow & & \searrow & \\
 E \times_B A & & & & E \times_B A \\
 & \searrow \exists! \langle \text{proj}_1, 1_{E \times_B A} \rangle & & \nearrow & \\
 & E \times_B (E \times_B A) & \longrightarrow & E \times_B A & \\
 & \downarrow \text{proj}_1, E \times_B (E \times_B A) & & \downarrow p(\text{proj}_1, E \times_B A) & \\
 \text{proj}_1, E \times_B A & \searrow & E & \xrightarrow{p} & B
 \end{array}$$

$$\circledast = (\text{pr}\ddot{o}\text{j}_2)(\langle \text{proj}_1, 1_{E \times_B A} \rangle)$$

In order to conclude we need to check some compositions so that, by the universal property of the pullback $E \times_B A$ in the right of the following diagram, we obtain the identity morphism.

$$\begin{array}{ccccccc}
 & & 1_{E \times_B A} & & \text{pr}\ddot{o}\text{j}_2 & & \\
 & \nearrow & & \searrow & & \nearrow & \\
 E \times_B A & \xrightarrow{\langle \text{proj}_1, 1_{E \times_B A} \rangle} & E \times_B (E \times_B A) & \longrightarrow & E \times_B A & \xrightarrow{\text{proj}_2, E \times_B A} & A \longleftarrow E \times_B A \\
 & \searrow & \downarrow \text{proj}_1, E \times_B (E \times_B A) & & \downarrow (\alpha)(\text{proj}_2, E \times_B A) & \swarrow \alpha & \\
 & & & & B & & \\
 & & & & \uparrow p & & \\
 & & & & E & & \\
 & \nearrow \text{proj}_1, E \times_B A & & \nearrow \text{proj}_1, E \times_B A & & \nearrow \text{proj}_1, E \times_B A &
 \end{array}$$

$$\circledast = (\text{pr}\ddot{o}\text{j}_2)(\langle \text{proj}_1, 1_{E \times_B A} \rangle) = \text{Id}_{(E \times_B A, \text{proj}_1)}$$

Let check the composing the projections of $E \times_B A$ with $(\text{pr}\ddot{o}\text{j}_2)(\langle \text{proj}_1, 1_{E \times_B A} \rangle)$ is the same of composing those projection with $\text{Id}_{(E \times_B A, \text{proj}_1)}$. Following the first composition

$$\begin{aligned}
 (\text{proj}_2, E \times_B A)(\text{pr}\ddot{o}\text{j}_2)(\langle \text{proj}_1, 1_{E \times_B A} \rangle) &= (\text{proj}_2, E \times_B A)(\text{proj}_2, E \times_B (E \times_B A))(\langle \text{proj}_1, 1_{E \times_B A} \rangle) = \\
 &= (\text{proj}_2, E \times_B A)(1_{E \times_B A}) = \text{proj}_2, E \times_B A = (\text{proj}_2, E \times_B A)(\text{Id}_{(E \times_B A, \text{proj}_1)})
 \end{aligned}$$

and the second

$$(\mathrm{proj}_{1,E \times_B A})(\tilde{\mathrm{proj}}_2)(\langle \mathrm{proj}_1, 1_{E \times_B A} \rangle) = (\mathrm{proj}_{1,E \times_B (E \times_B A)})(\langle \mathrm{proj}_1, 1_{E \times_B A} \rangle) =$$

$$\mathrm{proj}_{1,E \times_B A} = (\mathrm{proj}_{1,E \times_B A})(\mathrm{Id}_{(E \times_B A, \mathrm{proj}_1)})$$

Bibliografia

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